INCH POUNDS
A-A-59141B
07 JUL 2017
SUPERSEDING
A-A-59141A
25 JUL 2006

COMMERCIAL ITEM DESCRIPTION

TRUCK, LIFT, FORK, ELECTRIC, SOLID-TIRED, 2000 POUNDS CAPACITY AT 24 INCH LOAD CENTER, 144 INCH LIFT HEIGHT

The General Services Administration has authorized the use of this Commercial Item Description (CID) as a replacement for type "EE" of MIL-T-21869-35 for all federal agencies.

1. SCOPE. This commercial item description covers the general requirements for an electric, front wheel drive, rear wheel steer, solid-tired, 2,000 pound at 24 inch load-center capacity, forklift truck. It will be used for handling cargo in and around warehouses, loading platforms, and docks; on paved or other hard surfaces.

2. CLASSIFICATION. The truck shall be Type E, unless otherwise specified as Type EE or Type EX, in accordance with UL 583.

3. SALIENT CHARACTERISTICS:

3.1 Safety. The forklift shall comply with ANSI/ITSDF B56.1 and OSHA standards in effect at the time of manufacture. A seat belt conforming to SAE J386 and restraint devices designed to ensure the operator's upper body remains entirely within the protection of the overhead guard in the event of tip-over shall be provided. Unless otherwise specified (see 7.2), the final color shall be yellow for safe operation in an indoor environment. The truck shall be Type E, unless otherwise specified as Type EE or Type EX, in accordance with UL 583. A mounted fire extinguisher shall be provided.

3.2 Environmental requirements. The forklift shall operate as specified herein in any ambient temperature from 0 to 120 degrees F.

3.3 Battery. The battery shall have adequate capacity to provide not less than 7.0 hours of continuous operation under normal, industrial usage. The battery shall have "SB" type half connectors. When specified (see 7.2), the truck shall be shipped with the following items: battery and/or battery charger.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data which may improve this document should be sent to: U. S. AIR FORCE, AFLCMC/WN-SEV, Robins AFB GA 31098 or emailed to SPEC99@us.af.mil .

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3.4 Battery discharge indicator. A battery discharge indicator shall be located so that it is visible to the seated operator.

3.5 Controls. All controls shall be in accordance with ANSI/ITSDF B56.1. All load motion controls shall be self-centering, that is they shall return to the neutral position when released.

3.6 Hydraulic system. A pressure relief protection device is required along with pump(s), cylinders, control valves, filter(s), reservoir, hoses, and all other components necessary to make a complete hydraulic system.

3.7 Uprights and carriage. With no load on the forks, the seated operator shall be able to see at least one fork tip at all lift heights.

3.8 Forks. Fork dimensions shall be 1.56 inches maximum thickness, 4.06 inch maximum width, and 36.0 (+ or - 1.0) inch long. Fork thickness at the tip shall be not more than 0.5 inch, and there shall be a taper on the underside of the forks of not less than 14 inches.

3.9 Fork positioner. The forks shall be manually adjustable with spacing of 14.0 inches (or less) when in most closed position, and 30.0 inches (or more) when in most spread position. These dimensions shall be measured to the outside edges of the forks.

3.10 Side shift. Hydraulically powered side shifting shall be furnished. It shall be controlled by seated operator, with at least 4.0 inches side shift each side of center (8.0 inches minimum total travel) with rated load on the forks.

3.11 Load backrest. The load backrest, in conjunction with the forks and carriage, shall provide a vertical rear load guard of at least 48 inches high, measured from the load horizontal carrying surfaces of the forks.

3.12 Steering. Power steering shall be furnished.

3.13 Service brakes. Service brakes shall conform to ANSI/ITSDF B56.1.

3.14 Parking brake. Parking brake shall conform to ANSI/ITSDF B56.1.

3.15 Electrical system. All components necessary to make a complete electrical system shall be furnished. The truck shall be equipped with a back-up alarm.

3.16 Instruments. The instruments, gauges or warning indicator lights that are on the manufacturer's standard commercial model shall be furnished. In addition, an unlighted hour meter capable of totaling at least 9999 hours shall be included.

3.17 External lights. Minimum of two floodlights shall be furnished. One in front, facing forward, and one mounted in the rear, facing rearward.

3.18 Taillight. One combination stoplight and taillight shall be provided.

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3.19 Horn. A horn shall be furnished, with the horn actuator located in the center of the steering wheel.

3.20 Operator's overhead guard. The guard shall be in accordance with ANSI/ITSDF B56.1, including inside clearances. It shall also be a falling-object-protective-structure (FOPS) conforming to ANSI/ITSDF B56.1. The height, from the ground to the top of the overhead guard shall be not more than 86 inches.

3.21 Painting. The forklift shall be primed and painted using standard commercial practices. Unless otherwise specified (see 7.2), the final color shall be yellow.

3.22 Walkway coating. Step surfaces shall be made of, or covered with, a non-slip material.

3.23 Slinging and tie-down provisions. There shall be permanently installed slinging devices that enable the truck to be lifted in its normal travel position, and tie-downs that enable the truck to be tied down to the floor of the transportation media. All devices shall be capable of accepting a chain and hook with an outside diameter of 3.0 inches and shall have a structural safety factor of at least 2.0 to 1, based on static load. The notation “LIFT HERE” or “TIE DOWN” shall be stenciled, in black letters, near each lifting and tie-down device.

3.24 Identification plate. A corrosion resistant identification plate shall be permanently installed at an easily accessible place on the truck. It shall contain the following information:

NOMENCLATURE:
MANUFACTURER'S NAME:
MANUFACTURER'S ADDRESS:
MANUFACTURER'S SERVICE PHONE NUMBER:
MANUFACTURER'S MODEL NUMBER:
MANUFACTURER'S SERIAL NUMBER:
DATE OF MANUFACTURE:
CONTRACT NUMBER:
NATIONAL STOCK NUMBER:
REGISTRATION NUMBER:

3.25 Markings. All stencils, decals, plates, etc. normally provided on the manufacturer’s standard forklift shall be included. This shall include, as a minimum, the forklift’s capacity in pounds, and all information, cautions and warnings normally provided.

3.26 Performance. The truck shall be capable of meeting the following performance requirements:

(1) Capability to safely handle a 2,000 pound, 48 inch per side cube with the center of gravity at the center of the cube. This shall be defined, and referred to in this document, as the rated load. The forklift shall be able to safely handle this rated load at all lift heights and fork side shift positions that it is capable of attaining.

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(2) Load lifting speed: Shall be capable of raising rated load at a speed of at least 30 feet per minute.

(3) Load lowering speed: Shall be not more than 80 feet per minute with rated load of forks and not less than 20 feet per minute with no load on the forks.

(4) Right angle turn: With rated load on the forks and forklift located perpendicular to a wall, and with the front of the load against the wall, the truck shall be able to back up and make a complete right angle turn (ending up parallel to the wall) within 136 inches.

(5) Travel speed: Shall travel at a speed of at least 5 miles per hour, in both forward and reverse directions, while carrying rated load on the forks.

(6) Slope ascension: With the truck facing up the slope and with rated load on the forks it shall be able to accelerate up a 15 percent slope from a dead stop.

(7) Upright tilt: With no load on the truck the forks shall have at least 2.0 degrees forward tilt, and at least 6.0 degrees of rear tilt.

(8) Lowest upright height (collapsed mast height): With the unloaded forks fully lowered, the dimension from ground to top surface of the uprights, with uprights vertical, shall not exceed 86 inches.

(9) Fork lift height. At least 144 inches with rated load on forks and forks horizontal, when measured from ground to top surfaces of the forks.

(10) Free lift height. The height that the forks can raise the rated load before the collapsed mast height is increased, shall be at least 40 inches.

(11) Drift of load. With hydraulic fluid at ambient temperature, lift assembly shall hold rated load at maximum lift height for at least 10.0 minutes with not more than 1 to 3/4 inch of vertical drift, and not more than 1.0 degree of rotational drift.

(12) Stability. The forklift shall meet the ANSI/ITSDF B56.1 "forward stacking", "forward travel", "lateral stacking", and "lateral travel" stability requirements.

(13) Electromagnetic radiation. The truck shall comply with the EMI requirements shown in table I.

TABLE I. Electromagnetic interference (EMI)

Name	Description	Parameters
SAE J551-1	Compatibility	N/A
SAE J551-2	Emissions	(Frequency Range: 30 to 1000 MHz)
SAE J551-11	Susceptibility	(Frequency Range: 100 kHz to 18 GHz) 20 V/m (100 kHz to 200 MHz) 50 V/m (above 200 MHz)

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4. REGULATORY REQUIREMENTS.

4.1 The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR). However, used, rebuilt or remanufactured components, pieces, and parts shall not be incorporated in the forklift.

5. PRODUCT CONFORMANCE PROVISIONS.

5. 1 Product Conformance. The products provide shall meet all the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards and quality assurance practices, and shall be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

5.2 Market Acceptability. The following market acceptability criteria are necessary to document the quality of the product to be provided under this CID.

5.2.1 Commercial. The vehicle furnished shall be, as of the date of award, the latest model (with minor modifications, as that term is defined in DoD FAR Supplement 252.211-7012, Paragraph K-85 of the solicitation) of an established product previously produced and sold in substantial commercial quantities (exceeding the criteria at FAR 15.804-3(f)(2)(ii)(A)) for at least three years. The vehicle furnished, or earlier models of it, absent minor modifications, shall have routinely been supported with spare/repair parts which were produced or sold in the normal course of business. At the time of delivery, the contractor shall furnish the Administrative Contracting Officer with verification of compliance with these requirements. Offers shall provide, as part of their proposals, the following information on the vehicle (or its earlier models) being offered as a commercial item:

(1) Total sales to the U.S. Government or to contractors for U.S. Government use during the three years.

(2) Total sale of the item to the general public during the three years.

(3) Length of time the item has been sold in the commercial market place.

6. PACKAGING. Preservation, packing and marking shall be specified in the contract or order.

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7. NOTES.

7.1 Source of Documents.

7.1.1 ANSI/ITSDF documents may be obtained online at www.itsdf.org or from Industrial Truck Standards Development Foundation, Suite 460, 1750 K Street NW, Washington, DC 20006.

7.1.2 UL documents may be obtained online at www.shopulstandards.com or from UL Comm 2000, 151 Eastern Avenue, Bensenville, IL 60106.

7.1.3 SAE documents may be obtained online at www.sae.org or from SAE Inrenational, Inc., 400 Commonwealth Drive, Warrendale PA 15096.

7.1.4 Federal Acquisition Regulation (FAR) documents may be obtained online at www.acquisition.gov/far or at the Superintendent of Documents, P.O. Box 371954, Pittsburgh, PA 15250.

7.2 Ordering Data. When specified the following shall be furnished:

a. Final color other than yellow (see 3.21).
b. Additional battery (see 3.3)
c. Battery charger (see 3.3)

7.3 National Stock Numbers (NSNs). The following is a list of master NSNs assigned that correspond to this CID. The list is not indicative of all possible NSNs associated with this CID.

NSN	DOCUMENT NUMBER	DOCUMENT STATUS
3930-00-678-2580	MIL-T-21869-35	Cancelled
3930-01-455-5463	A-A-59141	Active

7.4 Subject term (key word) listing.

1. Electric powered forklift
2. Rear wheel steer forklift
3. Solid tired warehouse forklift
4 2000 Pound cargo handling

7.5 Changes from previous issue. The margins of this CID are marked with vertical lines to indicate where changes from the previous issue were made. This was done as convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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Preparing Activity:
Air Force – 184
(Project 3930-2017-006)

Reviewers:
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Air Force - 99
DLA - CQ

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Air Force – 99

Civil agencies:
GSA - FSS

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